SOC Lab: Introduction to Wazuh for Threat Detection & Monitoring

Course: Security Operations Center (SOC)
Module: SIEM & Endpoint Monitoring Fundamentals

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Lab Overview & Learning Objectives

Welcome, SOC Analysts! In this lab, you will build a foundational security monitoring system. You will deploy the Wazuh platform to gain visibility into endpoint activities and detect changes in your environment.

By the end of this lab, you will be able to:

- **Deploy** a central SIEM (Security Information and Event Management) manager.
- **Install and register** an endpoint agent on a Windows system.
- **Configure** File Integrity Monitoring (FIM) to detect file changes.

- **Verify** agent connectivity and health from a central dashboard.
- **Analyze** and **interpret** security alerts generated by monitored events.

Scenario: The Need for Visibility

You are a Level 1 SOC Analyst. The CISO is concerned about unauthorized changes to critical files on company workstations. Your task is to implement a solution that can:

1. Collect logs and events from a Windows 10 endpoint.
2. Monitor a specific folder (C:\Users\<YourName>\SensitiveFiles) for any creation, modification, or deletion of files.
3. Generate real-time alerts in a central dashboard for your team to investigate.

You have chosen the open-source Wazuh platform to accomplish this mission.

Lab Architecture

Component	Host	Role
Wazuh Manager	Ubuntu (VirtualBox)	The "brain" of the SOC. Collects, analyzes, and stores data from agents.
Wazuh Agent	Windows (Host Machine)	The "eyes and ears" on the endpoint. Sends logs and system events to the manager.

Network Configuration:

- Set your Ubuntu VM's network adapter to **Bridged Adapter**. This places it on the same network as your host machine, allowing them to communicate.

Prerequisites

- VirtualBox installed.
- Ubuntu Server 20.04+ installed in VirtualBox (with bridged networking enabled).
- Internet access on the Ubuntu VM.
- Administrative access on your Windows host machine.

Part 1: Deploying the Wazuh Manager (Ubuntu)

Run the following steps on your Ubuntu VirtualBox server via a terminal window.

1. **Add Wazuh GPG Key**

This command adds the official Wazuh GPG key to your system, which is used to verify the authenticity of the software packages.

bash

curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o /usr/share/keyrings/wazuharchive-keyring.gpg

```
function passwords_updateInternalUsers() {

    common_logger "Updating the internal users."
    backup_datetime=$(date +"%Y%m%d_%H%M%S")
    internal_users_backup_path="/etc/wazuh-indexer/internalusers-backup"
    passwords_getNetworkHost
    passwords_createBackUp

    eval "mkdir -p ${internal_users_backup_path} ${debug}"
    eval "cp /etc/wazuh-indexer/backup/internal_users.yml ${internal_users_backup_path}/internal_us
rs_${backup_datetime}.yml.bkp ${debug}"
    eval "chmod 750 ${internal_users_backup_path} ${debug}"
    eval "chmod 640 ${internal_users_backup_path}/internal_users_${backup_datetime}.yml.bkp"
    eval "chown -R wazuh-indexer:wazuh-indexer ${internal_users_backup_path} ${debug}"
    common_logger "A backup of the internal users has been saved in the /etc/wazuh-indexer/internal
sers-backup folder."

    eval "cp /etc/wazuh-indexer/backup/internal_users.yml /etc/wazuh-indexer/opensearch-security/in
ernal_users.yml ${debug}"
    eval "rm -rf /etc/wazuh-indexer/backup/ ${debug}"
    common_logger -d "The internal users have been updated before changing the passwords."

}

main "$@"
22/02/2026 15:12:49 INFO: Starting Wazuh installation assistant. Wazuh version: 4.12.0
22/02/2026 15:12:49 INFO: Verbose logging redirected to /var/log/wazuh-install.log
22/02/2026 15:13:05 ERROR: Wazuh manager already installed.
22/02/2026 15:13:05 INFO: If you want to overwrite the current installation, run this script adding
the option -o/--overwrite. This will erase all the existing configuration and data.
```

2. **Download and Execute Installation Script**

This script will automatically download and install all necessary Wazuh components (Manager, Indexer, and Dashboard).

bash curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && sudo bash ./wazuh-

install.sh -a -i

- -a: Installs **all** components.
- -i: Runs in **interactive** mode (follow any on-screen prompts).

SOC Note: This automated installation is perfect for a lab. In a production environment, these components might be installed on separate servers for scalability and security.

Wazuh Management Commands

sudo systemctl restart wazuh-manager

sudo systemctl stop wazuh-manager

sudo systemctl start wazuh-manager

sudo systemctl status wazuh-manager

```
root@icdfa-server: ~
Session  Actions  Edit  View  Help
● wazuh-manager.service - Wazuh manager
     Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; ven>
     Active: active (running) since Sun 2026-02-22 15:38:33 UTC; 6h ago
      Tasks: 154 (limit: 5185)
     Memory: 1.5G
     CGroup: /system.slice/wazuh-manager.service
             ├─1179 /var/ossec/framework/python/bin/python3 /var/ossec/api/s>
             ├─1180 /var/ossec/framework/python/bin/python3 /var/ossec/api/s>
             ├─1181 /var/ossec/framework/python/bin/python3 /var/ossec/api/s>
             ├─1183 /var/ossec/framework/python/bin/python3 /var/ossec/api/s>
             ├─1187 /var/ossec/framework/python/bin/python3 /var/ossec/api/s>
             ├─1245 /var/ossec/bin/wazuh-authd
             ├─1263 /var/ossec/bin/wazuh-db
             ├─1293 /var/ossec/bin/wazuh-execd
             ├─1309 /var/ossec/bin/wazuh-analysisd
             ├─1327 /var/ossec/bin/wazuh-syscheckd
             ├─1348 /var/ossec/bin/wazuh-remoted
             ├─1385 /var/ossec/bin/wazuh-logcollector
             ├─1408 /var/ossec/bin/wazuh-monitord
             └─1448 /var/ossec/bin/wazuh-modulesd

Feb 22 15:38:27 icdfa-server env[715]: wazuh-logcollector: Process 70279 not>
Feb 22 15:38:28 icdfa-server env[715]: Started wazuh-logcollector...
Feb 22 15:38:28 icdfa-server env[715]: wazuh-monitord: Process 70309 not use>
Feb 22 15:38:29 icdfa-server env[715]: Started wazuh-monitord...
Feb 22 15:38:29 icdfa-server env[715]: wazuh-modulesd: Process 70366 not use>
Feb 22 15:38:30 icdfa-server env[1445]: 2026/02/22 15:38:30 wazuh-modulesd:r>
lines 1-27
```

Part 2: Accessing the Wazuh Dashboard

After the installation completes successfully:

1. Find your Ubuntu VM's IP address: ip addr show
2. open a web browser and navigate to:
   https://<UBUNTU_VM_IP_ADDRESS>

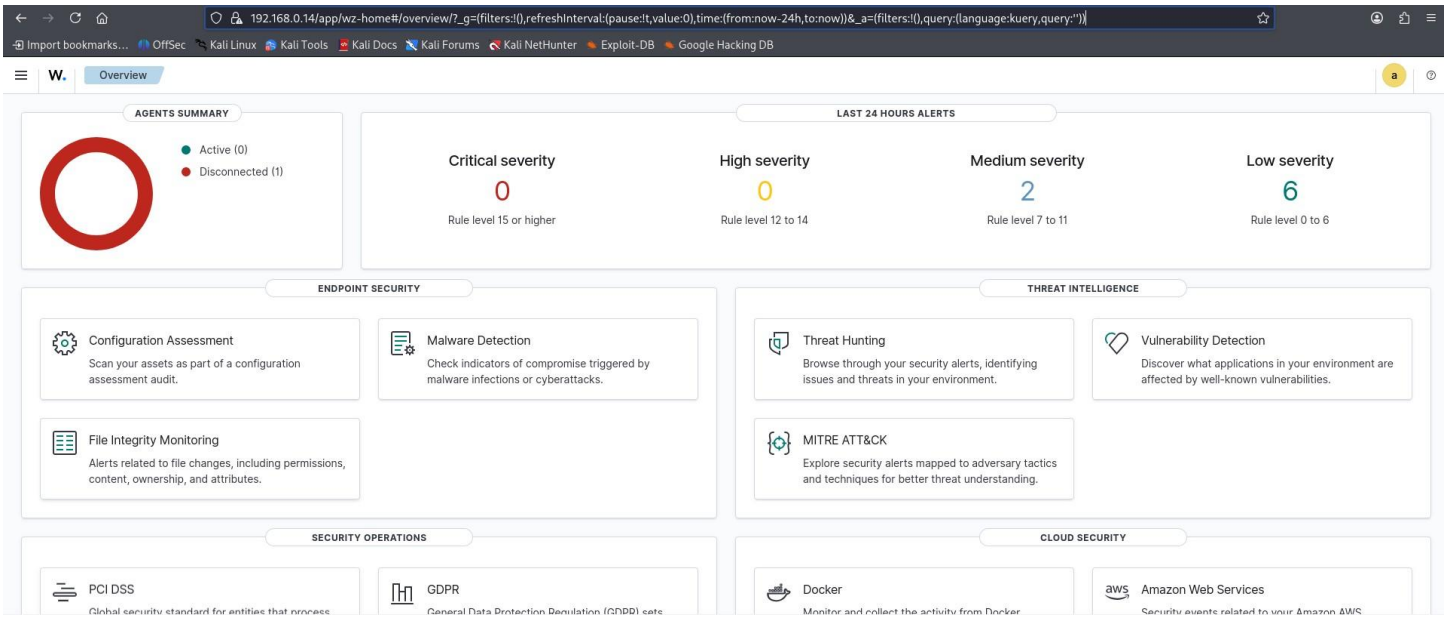
```
icdfa-server@icdfa-server:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.0.14  netmask 255.255.255.0  broadcast 192.168.0.255
        inet6 fe80::a00:27ff:fe36:a2a7  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:36:a2:a7  txqueuelen 1000  (Ethernet)
        RX packets 80  bytes 38260 (38.2 KB)
        RX errors 3  dropped 0  overruns 0  frame 3
        TX packets 91  bytes 11673 (11.6 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 244  bytes 15608 (15.6 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 244  bytes 15608 (15.6 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

icdfa-server@icdfa-server:~$ [  160.397916] cloud-init[3225]: Cloud-init v. 23.1.2-0ubuntu0~20.04.2
running 'modules:final' at Sun, 22 Feb 2026 15:40:06 +0000. Up 159.75 seconds.
[  160.402541] cloud-init[3225]: Cloud-init v. 23.1.2-0ubuntu0~20.04.2 finished at Sun, 22 Feb 2026
15:40:07 +0000. Datasource DataSourceNone.  Up 160.36 seconds
[  160.407336] cloud-init[3225]: 2026-02-22 15:40:07,108 - cc_final_message.py[WARNING]: Used fallba
ck datasource
```

3. **Accept the browser security warning.** This is expected because we are using a self-signed certificate for this lab.

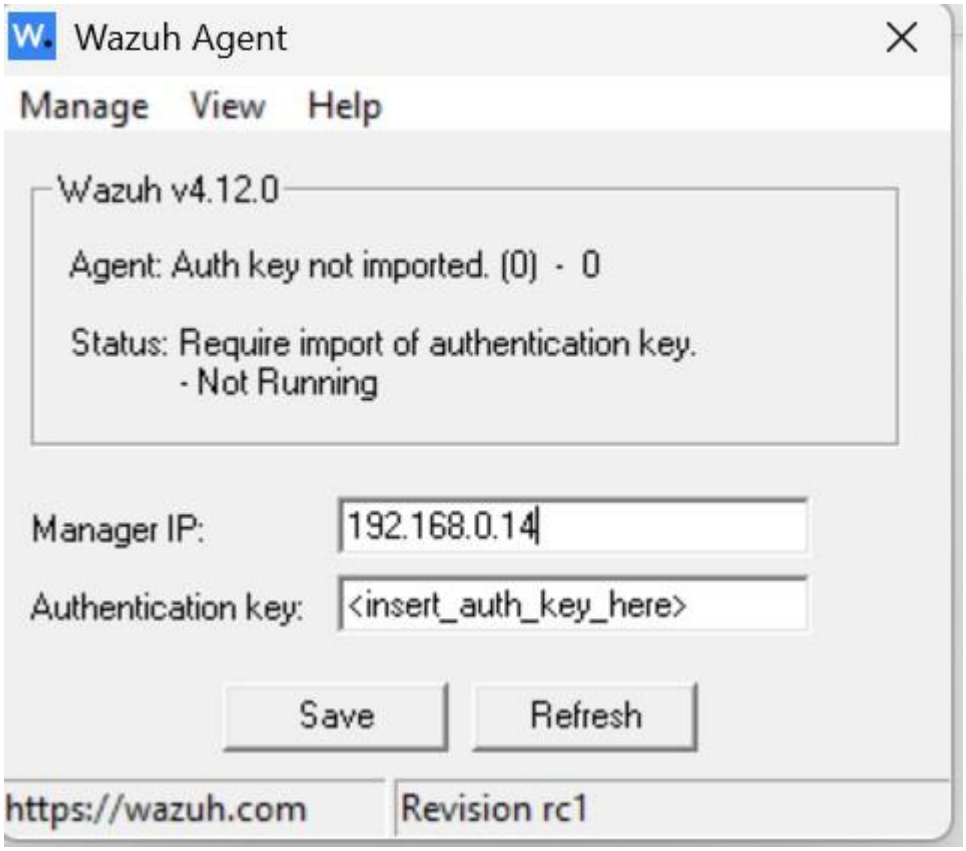
4. Log in using the username and password provided at the end of the installation script output (check your terminal). The default username is admin.



SOC Note: This dashboard is your **SOC Primary Console**. From here, you will monitor the health of your agents and investigate any alerts that may arise.

Part 3: Installing the Wazuh Agent (Windows)

1. On your **Windows host**, download the latest Wazh Agent MSI installer from the official page: https://packages.wazuh.com/4.x/windows/wazuh-agent-4.12.0-1.msi 2. Run the downloaded MSI file and install it with the default settings.



Part 4: Agent-Manager Registration

For the agent to communicate with the manager, they must share a unique authentication key.

1. Generate Agent Key on Manager

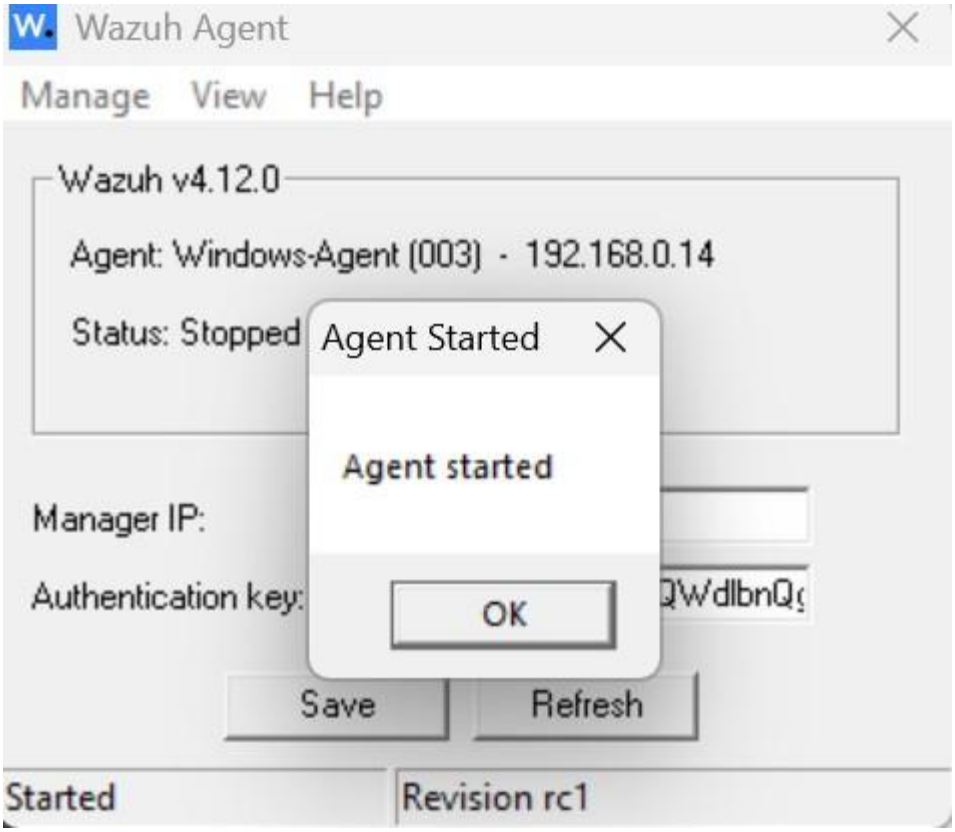
On your **Ubuntu VM terminal**, run the agent management

tool: bash sudo /var/ossec/bin/manage_agents

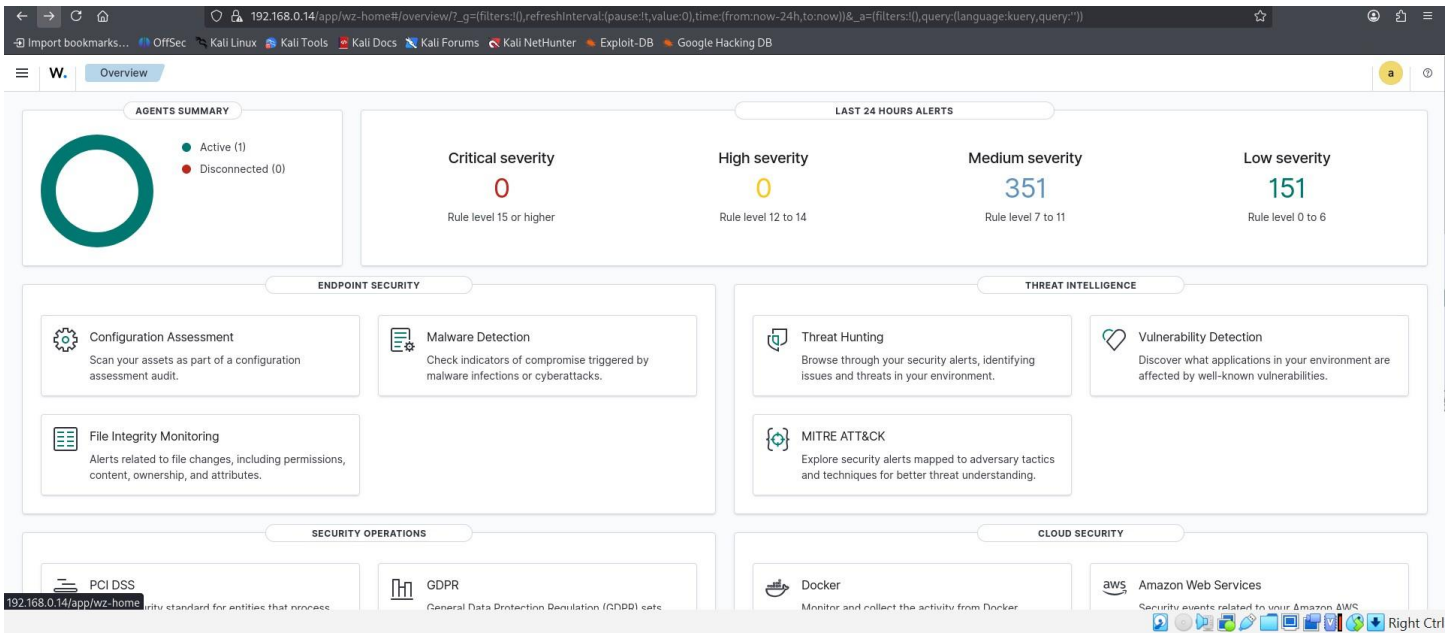
- Press A to **Add an agent**.
- Enter a name (e.g., Windows-SOC-Lab).
- For the IP address, you can leave it as any (or enter your Windows host's IP if it's static).
- Confirm to add the agent.
- Press E to **Extract the key** for the agent you just created.
- **Carefully copy the long generated key** to your clipboard. You will need it for the next step.

2. Apply Key in the Windows Agent

1. On your **Windows host**, open the **Wazuh Agent Manager** from the Start Menu.
2. Go to the **Management** tab.
3. Paste the key you copied from the Ubuntu manager into the "Agent key" field.
4. In the "Manager address" field, enter the IP address of your **Ubuntu VM**.
5. Click **Apply** and then **Restart** the agent service.



Verification: Return to the Wazuh Dashboard in your browser. Navigate to **Agents**. Within a minute, you should see your Windows-SOC-Lab agent with a green dot and status **Active**.



Part 5: Implementing File Integrity Monitoring (FIM)

We will now configure the agent to monitor a folder for any changes—a critical control for detecting unauthorized activity.

1. Edit Agent Configuration

1. On your **Windows host**, navigate to C:\Program Files (x86)\ossec-agent\.

2. Open the ossec.conf file in a text editor like Notepad++ or Notepad (as Administrator).

```
  </sca>

  <!-- File integrity monitoring -->
  <syscheck>

    <disabled>no</disabled>

    <!-- Frequency that syscheck is executed default every 12 hours -->
    <frequency>43200</frequency>

<directories realtime="yes">C:\Users\Administrator
\SensitiveFiles</directories>

    <!-- Default files to be monitored. -->
    <directories recursion_level="0" restrict="regedit.exe$|system.ini
$|win.ini$">%WINDIR%</directories>

    <directories recursion_level="0" restrict="at.exe$|attrib.exe$|cacls.exe
$|cmd.exe$|eventcreate.exe$|ftp.exe$|lsass.exe$|net.exe$|net1.exe$|netsh.exe
$|reg.exe
$|regedt32.exe|regsvr32.exe|runas.exe|sc.exe|schtasks.exe|sethc.exe|subst.ex
e$">%WINDIR%\SysNative</directories>
```

3. Find the <syscheck> section.

4. Inside this section, add the following line to monitor a folder in real-time. **Replace** abc **with your Windows username or create a different test folder.**

xml

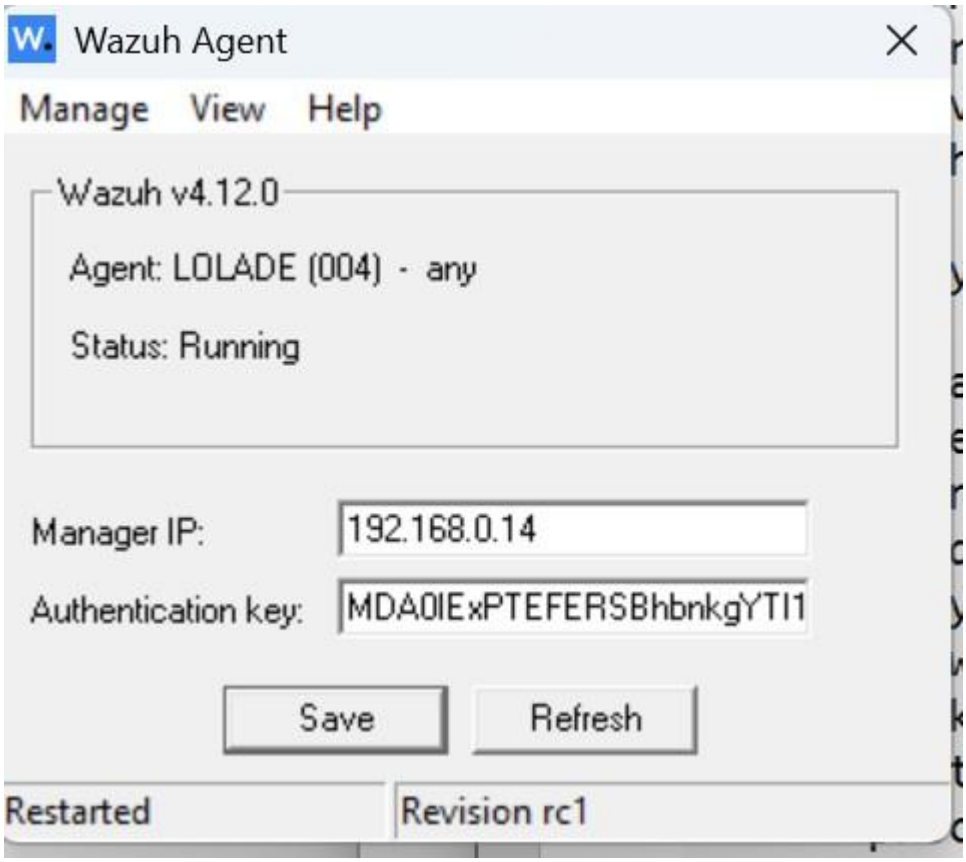
<directories realtime="yes">C:\Users\AMINU IDRIS\SensitiveFiles </directories>

```
File    Edit    View
    </synchronization>
<directories realtime="yes">C:\Users\Administrator
\SensitiveFiles</directories>
  </syscheck>

  <!-- System inventory -->
  <wodle name="syscollector">
    <disabled>no</disabled>
    <interval>1h</interval>
    <scan_on_start>yes</scan_on_start>
    <hardware>yes</hardware>
    <os>yes</os>
    <network>yes</network>
    <packages>yes</packages>
    <ports all="no">yes</ports>
    <processes>yes</processes>

    <!-- Database synchronization settings -->
    <synchronization>
      <max_eps>10</max_eps>
    </synchronization>
  </wodle>
```

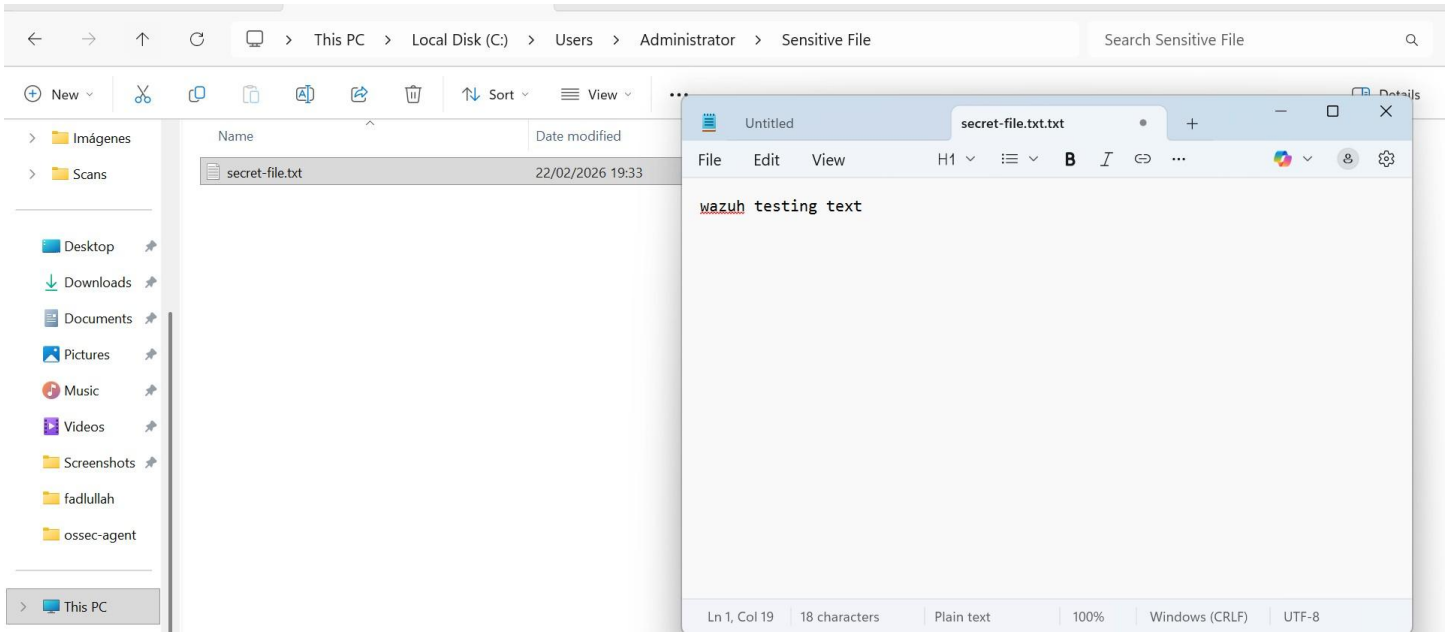
2. Restart the Agent



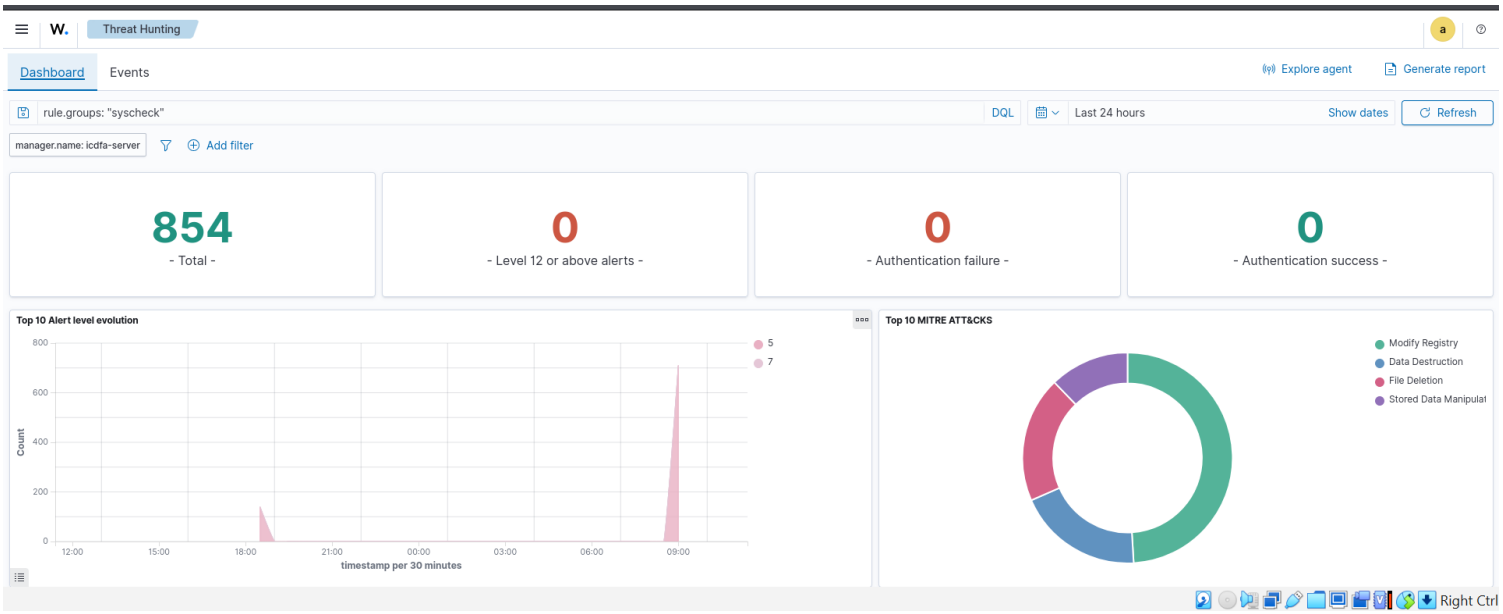
Part 6: SOC Verification & Alert Analysis

This is where you put on your SOC analyst hat.

1. **Create the Monitored Folder:** Create the folder C:\Users\abc\SensitiveFiles on your Windows machine.
2. **Generate Test Events:** Perform the following actions inside the SensitiveFiles folder:
    - Create a new text file (secret_plan.txt).
    - Modify the file by adding some text and saving it.
    - Delete the file.



3. **Investigate in the Dashboard:**
    - Go to the Wazuh Dashboard.
    - Navigate to **Security Events** (or use the search bar). You should see new alerts appearing in realtime.
    - Use the **Filters** on the left. Try selecting rule.groups: "syscheck" to see only File Integrity alerts.
    - Click on an alert to see the detailed information: What file was changed? What was the action (added, modified, deleted)? What was the timestamp?


Threat Hunting
Dashboard
Events
Explore agent
Generate report
rule.groups: "syscheck"
DQL
Last 24 hours
Show dates
Refresh
manager.name: icdfa-server
Add filter
854
- Total -
0
- Level 12 or above alerts -
0
- Authentication failure -
0
- Authentication success -
Top 10 Alert level evolution
Count
timestamp per 30 minutes
5
7
Top 10 MITRE ATT&CKS
Modify Registry
Data Destruction
File Deletion
Stored Data Manipulat
Right Ctrl